

Lifting Line Theory

Finite-wing Aerodynamics
(Downwash, Effective AoA, Induced Drag)



Tools needed for finite wing analysis

1. Curved vortex filament
2. Biot-Savart law
3. Helmholtz Vortex Theorems



Method of Analysis

Prandtl's Lifting Line Theory

Assumptions:

- 3-D steady potential flow (Inviscid, $\mu = 0$, Irrotational)
- Incompressible
- High Aspect Ratio ($AR > 6$)
- Low sweep
- Small crossflow (along span)
 - The flow looks like 2-D flow locally with α adjusted

Outputs:

- Lift distribution along the span
- Total lift and induced drag estimates
- Rolling moment
- Basic scaling of C_{Di} with C_L and geometry

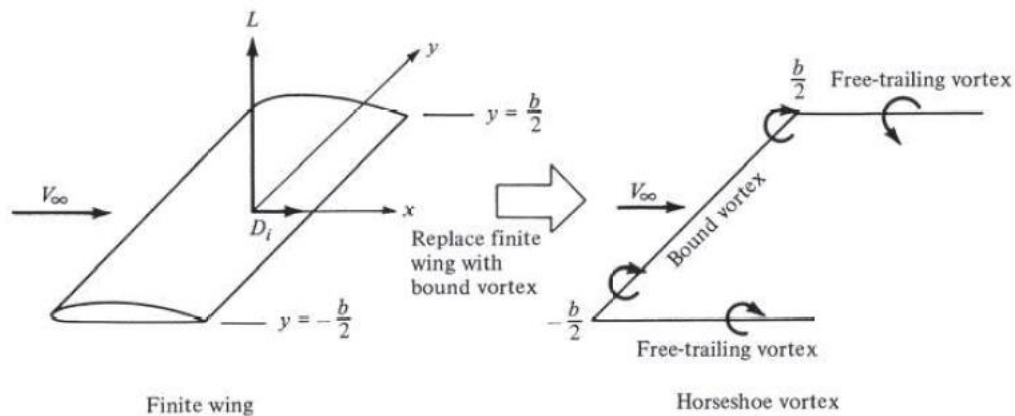
In reality, tip vortices induce downwash over the foil changing the effective angle of attack (α_{eff}) and adding a new drag called **Induced drag**.

The main goal is to predict the finite wing losses which are successfully done by estimating a foil with trailing edge vortices as a horseshoe vortex with no surface.

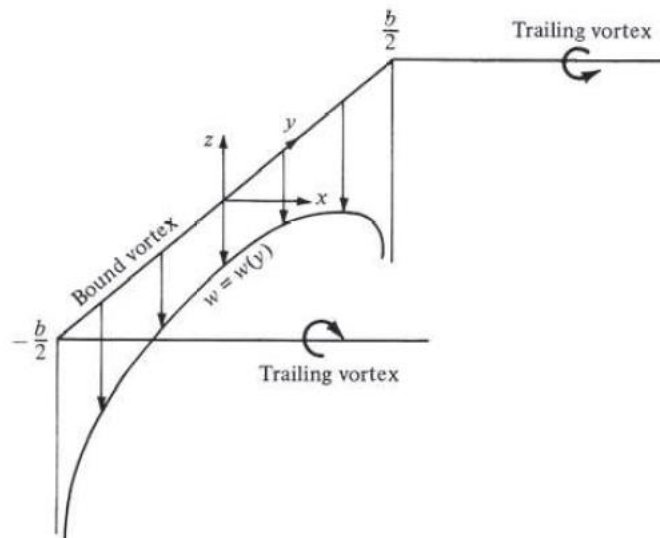
LLT is a combination of

- Biot – Savart Law
- Elementary flows
- Kutta – Joukowski Theorem

Replacement of the finite wing with a bound vortex:



Downwash distribution along the y-axis for a single horseshoe vortex:



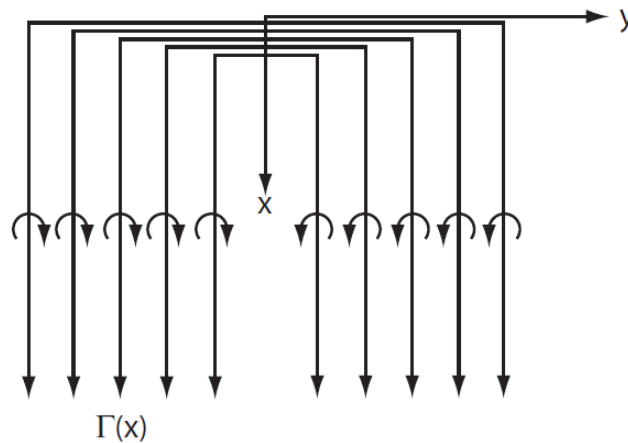
- If the origin is taken at the center of the bound vortex, then the velocity at any point y along the bound vortex induced by the trailing semi-infinite vortices is

$$w(y) = -\frac{\Gamma}{4\pi\left(\frac{b}{2} + y\right)} - \frac{\Gamma}{4\pi\left(\frac{b}{2} - y\right)}$$

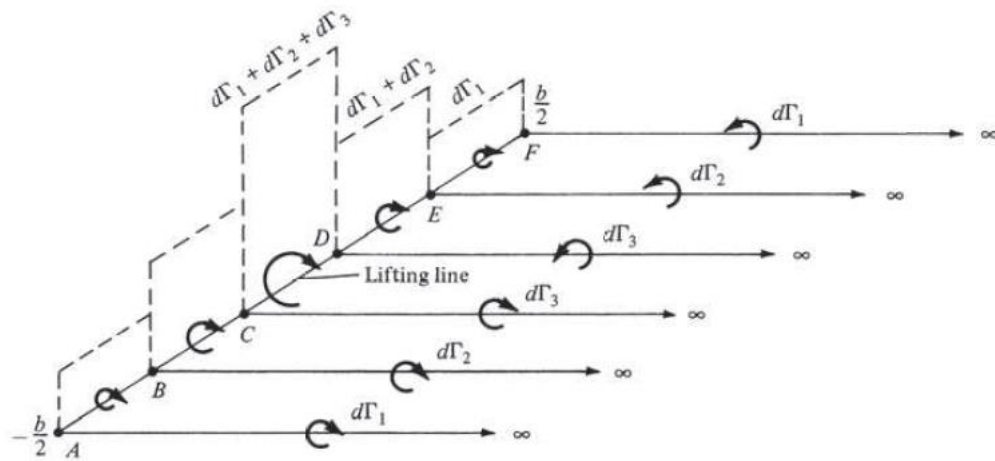
- The first term on the right-hand side is the contribution from the left trailing vortex (trailing from $-b/2$), and the second term is the contribution from the right trailing vortex (trailing from $b/2$).
- The $w(y)$ can be written as,

$$w(y) = -\frac{\Gamma}{4\pi} \frac{b}{(b/2)^2 - y^2}$$

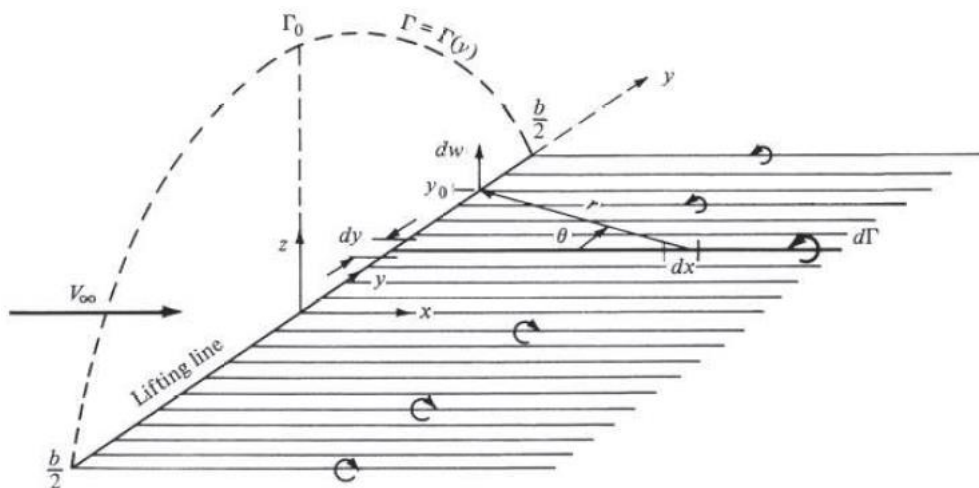
- Note the $w(y)$ approaches infinity when y approaches $b/2$. To overcome this problem, add multiple horseshoe vortices as shown in Figure below.



Superposition of a finite number of horseshoe vortices along the lifting line:



Superposition of an infinite number of horseshoe vortices along the lifting line:



From Biot – Savart Law,

$$\square \quad dV = \frac{-(d\Gamma/dy)dy}{4\pi(y_0 - y)}$$

Adding up all the segments, we get

$$\square \quad V = -\frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)dy}{y_0 - y}$$

Induced angle of attack:

$$\square \quad \alpha_{\text{induced}} = \tan^{-1}(-V/U_{\infty}) \approx -\frac{V}{U_{\infty}} \text{ (small angle approximation)}$$

$$\square \quad \alpha_{\text{induced}} = \frac{1}{4\pi U_{\infty}} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)dy}{y_0 - y} \text{ (Induced AoA due to } \Gamma)$$

Effective angle of attack:

$$\square \quad \alpha_{\text{eff}} = \alpha_{\text{geometry}} - \alpha_{\text{induced}}$$

Recall from Thin-airfoil theory,

$$\square \quad \frac{dC_l}{d\alpha} = 2\pi \Rightarrow dC_l = 2\pi d\alpha \Rightarrow C_l = 2\pi(\alpha_{\text{eff}} - \alpha_{L=0})$$

Lift equation,

$$\square \quad L' = \frac{1}{2} \rho U_{\infty}^2 c C_l \Rightarrow C_l = \frac{2\Gamma}{U_{\infty} c} = 2\pi(\alpha_{\text{eff}} - \alpha_{L=0})$$

$$\square \quad \alpha_{\text{eff}} = \frac{\Gamma}{\pi U_{\infty} c} + \alpha_{L=0}$$

From the above equations, we know that

$$\alpha_{\text{eff}} \rightarrow f(\Gamma)$$

$$\alpha_{\text{induced}} \rightarrow f(\Gamma)$$

Note: Only Γ is unknown!!

We use a Fourier series to solve this.

$$\square \quad \Gamma = 2bU_{\infty} \sum_{n=1}^N A_n \sin n\theta, \text{ where } y = -\frac{b}{2} \cos \theta$$

But angle of attack is a known parameter. Therefore,

$$\square \quad \alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_{\infty} c(y_0)} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_{\infty}} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)dy}{y_0 - y}$$

$\alpha \rightarrow$ function (y) – Geometric twist

$c \rightarrow$ function (y) – Chord variation

$\alpha_{L=0} \rightarrow$ function (y) – Aerodynamic twist

The fundamental equation of Prandtl's LLT:

$$\alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_\infty c(y_0)} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy) dy}{y_0 - y}$$

Once $\Gamma(y)$ is known, we get

- **Lift distribution**, $L' = \rho U_\infty \Gamma(y)$
- **Total lift**, $L = \int_{-b/2}^{b/2} L'(y) dy = \rho U_\infty \int_{-b/2}^{b/2} \Gamma(y) dy$
- **Induced drag**, $D_i' = L' \sin \alpha_i \approx L' \alpha_i \Rightarrow D_i = \int_{-b/2}^{b/2} L' \alpha_i dy = \rho U_\infty \int_{-b/2}^{b/2} \Gamma(y) \alpha_{\text{induced}} dy$

Important Outcome(s):

Induced angle of attack,

- $\alpha_{\text{induced}} = \frac{C_L}{\pi AR}$

Induced drag coefficient,

- $C_{Di} = \frac{C_L^2}{\pi AR}$ (for elliptical Γ distribution)
- **It is the most efficient distribution with lowest induced drag coefficient.**

In general,

1. Assume $\Gamma(y)$ – Elliptical distribution
2. From Γ solve for α_{induced}
3. From α_{induced} solve for α_{eff}
4. From α_{eff} estimate lift coefficient per unit span
5. Derive the lift per unit from lift coefficient and $\Gamma(y)$ using Kutta-Joukowski Theorem.
6. Repeat 1 – 5 steps till $\Gamma(y) = \Gamma(y)$ assumed.

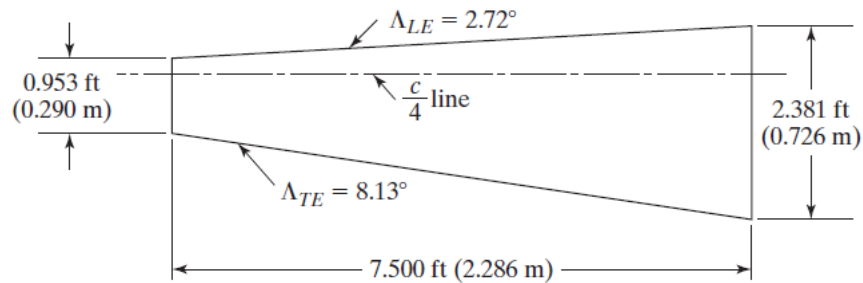
Classical LLT is not good for:

1. Low AR wings ($1 < AR < 5$)
2. Highly swept wing
3. Delta wing

This can be solved by using Lifting surface/Vortex Lattice method.

Key Results using LLT:

- Comparison of the theoretical and the experimental lift coefficients for an unswept wing in a subsonic stream.



- Planform for an unswept wing, $AR = 9.00$, Taper ratio = 0.40, airfoil section NACA 65–210.

